



nomic-ai/nomic-embed-text-v2-moe



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Sentence Similarity



sentence-transformers



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101 languages

nomic_bert

feature-extraction

custom_code



text-embeddings-inference

arxiv:2502.07972

arxiv:2205.13147



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Use this model



Model card



Files



xet



Community

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nomic-embed-text-v2-moe: Multilingual Mixture of Experts Text Embeddings

[Blog](#) | [Technical Report](#) | [AWS SageMaker](#) | [Atlas Embedding and Unstructured Data Analytics Platform](#)

This model was presented in the paper [Training Sparse Mixture Of Experts Text Embedding Models](#).

Model Overview

nomic-embed-text-v2-moe is a SoTA multilingual MoE text embedding model that excels at multilingual retrieval:

- **High Performance:** SoTA Multilingual performance compared to ~300M parameter models, competitive with models 2x in size
- **Multilinguality:** Supports ~100 languages and trained on over 1.6B pairs
- **Flexible Embedding Dimension:** Trained with [Matryoshka Embeddings](#) with 3x reductions in

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2,235,464



Safetensors



Model size 0.5B params

Tensor type F32

Files info

Inference Providers

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Sentence Similarity

This model isn't deployed by any Inference Provider.



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Ask for provider support

Model tree for nomic-ai/nomic-em...

Base model [FacebookAI/xlm-roberta-...](#)

Finetuned [nomic-ai/nomic-xlm...](#)Finetuned [nomic-ai/nomic-...](#)Finetuned (3) [this model](#)

Finetunes 27 models

Quantizations 18 models

storage cost with minimal performance degradations

- **Fully Open-Source:** Model weights, code, and training data (see code repo) released

Model	Params (M)	Emb Dim	BEIR	MIRACL	Pret rain Data	
Nomic Embed v2	305	768	52.86	65.80	✓	
mE5 Base	278	768	48.88	62.30	✗	
mGTE Base	305	768	51.10	63.40	✗	
Arctic Embed v2 Base	305	768	55.40	59.90	✗	
BGE M3	568	1024	48.80	69.20	✗	
Arctic Embed v2 Large	568	1024	55.65	66.00	✗	
mE5 Large	560	1024	51.40	66.50	✗	

Model Architecture

- **Total Parameters:** 475M
- **Active Parameters During Inference:** 305M

Spaces using nomic-ai/nomic-... 23

mteb/leaderboard

hakari-bench/leaderboard

GabeA/chatbench-leaderboard

pimcore/local-inference-service

akadhim/OncoScout

V1cky1611/research-agent

alexis07/ulysses

denniscraandijk/mtem-pruner

+ 15 Spaces

Collection including nomic-ai/nom...

Nomic Embed v2 Collection
Multilingu... • 5 items • Updat... • △ 24

Papers for nomic-ai/nomic-embed...

Training Sparse Mixture Of Experts ...
Paper • 2502.07972 • Publi... • △ 12

Matryoshka Representation Learning
Paper • 2205.13147 • Publi... • △ 30

- **Architecture Type:** Mixture of Experts (MoE)
- **MoE Configuration:** 8 experts with top-2 routing
- **Embedding Dimensions:** Supports flexible dimension from 768 to 256 through Matryoshka representation learning
- **Maximum Sequence Length:** 512 tokens
- **Languages:** Supports dozens of languages (see Performance section)

Paper Abstract

Transformer-based text embedding models have improved their performance on benchmarks like MIRACL and BEIR by increasing their parameter counts. However, this scaling approach introduces significant deployment challenges, including increased inference latency and memory usage. These challenges are particularly severe in retrieval-augmented generation (RAG) applications, where large models' increased memory requirements constrain dataset ingestion capacity, and their higher latency directly impacts query-time performance. While causal language models have addressed similar efficiency challenges using Mixture of Experts (MoE) architectures, this approach hasn't been successfully adapted to the general text embedding setting. In this paper, we introduce Nomic Embed v2, the first general purpose MoE text embedding model. Our model outperforms models in the same parameter class on both monolingual and multilingual benchmarks while also maintaining competitive performance with models twice its size. We open-source all code,

models, and evaluation data to ensure full reproducibility of our training pipeline at <https://github.com/nomic-ai/contrastors>.

Usage Guide

Installation

The model can be used through SentenceTransformers and Transformers.

For best performance on GPU, please install

```
pip install torch transformers einops git+ht
```

Important! The text prompt *must* include a *task instruction prefix*, instructing the model which task is being performed.

Please use `search_query:` before your queries/questions, and `search_document:` before your documents.

Transformers

If using Transformers, **make sure to prepend the task instruction prefix**.

```
import torch
import torch.nn.functional as F
from transformers import AutoTokenizer, AutoModel

tokenizer = AutoTokenizer.from_pretrained("nomic-ai/contrastors")
model = AutoModel.from_pretrained("nomic-ai/contrastors")

sentences = ['search_document: Hello!', 'search_query: Hello!']
```

```

def mean_pooling(model_output, attention_mask):
    token_embeddings = model_output[0]
    input_mask_expanded = attention_mask.unsqueeze(-1)
    return torch.sum(token_embeddings * input_mask_expanded, dim=-1)

encoded_input = tokenizer(sentences, padding='max_length',
                           max_length=128, return_tensors='pt')

model.eval()
with torch.no_grad():
    model_output = model(**encoded_input)
    embeddings = mean_pooling(model_output, encoded_input['attention_mask'])
    embeddings = F.normalize(embeddings, p=2, dim=-1)
    print(embeddings.shape)
    # torch.Size([2, 768])

    similarity = F.cosine_similarity(embeddings[0], embeddings[1])
    print(similarity)
    # tensor(0.9118)

```

For truncation, you can truncate before applying normalization

```

+ embeddings = embeddings[:, :matryoshka_dim]
embeddings = F.normalize(embeddings, p=2, dim=-1)

```

SentenceTransformers

With SentenceTransformers, you can specify the prompt_name as either "query" or "passage", and the task instruction will be included automatically.

```

from sentence_transformers import SentenceTransformer

model = SentenceTransformer("nomic-ai/nomic-embed-text-v1.5")
sentences = ["Hello!", ";Hola!"]
embeddings = model.encode(sentences, prompt_name='query')
print(embeddings.shape)
# (2, 768)

similarity = model.similarity(embeddings[0], embeddings[1])

```

```
print(similarity)
# tensor([[0.9118]])
```

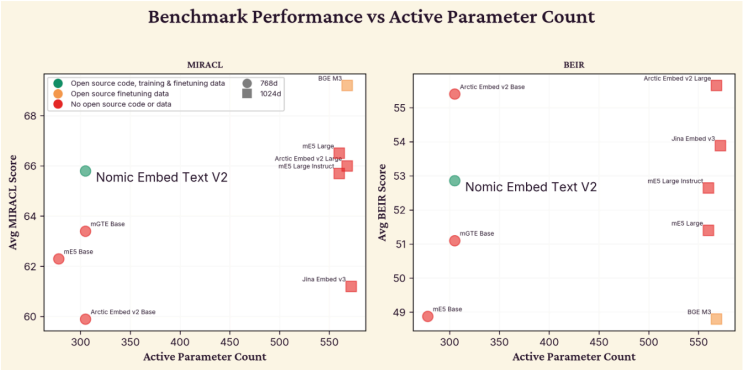
For truncation/Matryoshka embeddings, you can specify truncate_dim and use the model similarly

```
model = SentenceTransformer("nomic-ai/nomic-embed-text-v2-moe")
...

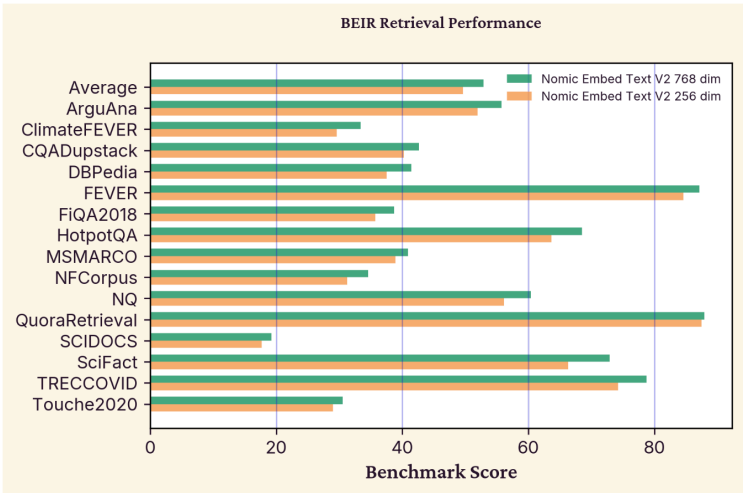
```

Performance

nomic-embed-text-v2-moe performance on BEIR and MIRACL compared to other open-weights embedding models:



nomic-embed-text-v2-moe performance on BEIR at 768 dimension and truncated to 256 dimensions:



Best Practices

- Add appropriate prefixes to your text:
 - For queries: "search_query: "
 - For documents: "search_document: "
- Maximum input length is 512 tokens
- For optimal efficiency, consider using the 256-dimension embeddings if storage/compute is a concern

Limitations

- Performance may vary across different languages
- Resource requirements may be higher than traditional dense models due to MoE architecture
- Must use `trust_remote_code=True` when loading the model to use our custom architecture implementation

Training Details

Code	Language	Pairs	Code	Language	Pairs	Code	Language	Pairs
en	English	234,553,344	be	Belarusian	589,824	my	Burmese	147,456
es	Spanish	210,010,112	ml	Malayalam	557,056	km	Khmer	131,072
fr	French	172,769,280	kn	Kannada	524,288	mg	Malagasy	131,072
de	German	169,426,944	mk	Macedonian	425,984	pa	Punjabi	131,072
it	Italian	104,251,392	ur	Urdu	409,600	ru-Latn	Russian (Latin)	131,072
pt	Portuguese	87,982,080	fy	Frisian	393,216	sn	Shona	131,072
pl	Polish	63,209,472	fil	Filipino	360,448	zh-Latn	Chinese (Latin)	131,072
nl	Dutch	50,118,656	te	Telugu	360,448	ha	Hausa	98,304
tr	Turkish	49,053,696	eu	Basque	344,064	he	Hebrew	98,304
ja	Japanese	43,433,984	sw	Swahili	327,680	hmn	Hmong	98,304
vi	Vietnamese	40,058,880	so	Somali	294,912	ht	Haitian	98,304
ru	Russian	38,731,776	sd	Sindhi	262,144	ja-Latn	Japanese (Latin)	98,304
id	Indonesian	36,470,784	uz	Uzbek	262,144	su	Sundanese	98,304
ar	Arabic	33,800,192	co	Corsican	245,760	bg-Latn	Bulgarian (Latin)	65,536
cs	Czech	29,966,336	hr	Croatian	245,760	gd	Scots Gaelic	65,536
ro	Romanian	24,772,608	gu	Gujarati	229,376	ny	Nyanja	65,536
sv	Swedish	24,608,768	hi-Latn	Hindi (Latin)	229,376	ps	Pashto	65,536
el	Greek	22,773,760	ceb	Cebuano	196,608	ku	Kurdish	49,152
uk	Ukrainian	19,841,024	eo	Esperanto	196,608	sh	Serbo-Croatian	49,152
zh	Chinese	18,661,376	jv	Javanese	196,608	am	Amharic	32,768
hu	Hungarian	18,448,384	la	Latin	196,608	ig	Igbo	32,768
da	Danish	14,548,992	zu	Zulu	196,608	lo	Lao	32,768
no	Norwegian	12,812,288	mn	Mongolian	180,224	mi	Maori	32,768
hi	Hindi	12,713,984	si	Sinhala	180,224	nn	Norwegian Nynorsk	32,768
fi	Finnish	12,697,600	el-Latn	Greek (Latin)	163,840	sm	Samoan	32,768
bg	Bulgarian	12,042,240	ga	Irish	163,840	yi	Yiddish	32,768
ko	Korean	10,354,688	ky	Kyrgyz	163,840	st	Sotho	16,384
sk	Slovak	8,962,048	tg	Tajik	163,840	tl	Tagalog	16,384
th	Thai	7,602,176				xh	Xhosa	16,384
iw	Hebrew	5,783,552				yo	Yoruba	16,384
ca	Catalan	5,701,632						
lt	Lithuanian	5,242,880						
fa	Persian	5,177,344						
ms	Malay	4,325,376						
sl	Slovenian	4,259,840						
lv	Latvian	3,211,264						
mr	Marathi	2,588,672						
bn	Bengali	2,457,600						
sq	Albanian	2,113,536						
cy	Welsh	2,048,000						

- Trained on 1.6 billion high-quality pairs across multiple languages
- Uses consistency filtering to ensure high-quality training data
- Incorporates Matryoshka representation learning for dimension flexibility
- Training includes both weakly-supervised contrastive pretraining and supervised finetuning

For more details, please check out the [blog.post](#) and [technical report](#).

Join the Nomic Community

- Nomic: <https://nomic.ai>
- Discord: <https://discord.gg/myY5YDR8z8>
- Twitter: https://twitter.com/nomic_ai

Citation


```
@misc{nussbaum2025trainingsparsemixtureexper
  title={Training Sparse Mixture Of Expe:
  author={Zach Nussbaum and Brandon Dude:
  year={2025},
  eprint={2502.07972},
  archivePrefix={arXiv},
  primaryClass={cs.CL},
  url={https://arxiv.org/abs/2502.07972}
}
```